Skills Summary

Factor Structure Puzzles

Use this reference while completing your worksheet or when studying.

Skill 1 — Finding the partner in a factor pair

Rule: Factors come in pairs. If one factor is known, its partner is the number divided by it.

number $\div$ known factor $=$ partner factor

The two sides of a rectangle made from unit squares are always a factor pair of its area.

Example: Fill in the missing factor: $8 \times$ ____ $= 104$
Step 1. One factor and the product are known, so the missing factor is the partner — divide the product by the factor I have.
Step 2. $104 \div 8 = 13$, and checking, $8 \times 13 = 104$. Missing factor: **13**

Try it: Fill in the missing factor: $6 \times$ ____ $= 84$

check: **14**

Skill 2 — Digit-sum tests for 3 and for 9

Rules: Add the digits.

divisible by 3 $\Leftrightarrow$ the digit sum is a multiple of 3   divisible by 9 $\Leftrightarrow$ the digit sum is a multiple of 9

divisible by 2 $\Leftrightarrow$ the last digit is even   divisible by 6 $\Leftrightarrow$ it passes the test for 2 *and* the test for 3

Example: Which digit goes in the box so that $3\square5$ is divisible by 9?
Step 1. Divisibility by 9 depends only on the digit sum, so I add the digits I know and see how far the sum is from a multiple of 9.
Step 2. $3 + 5 = 8$, and the next multiple of 9 is 9 itself, so the missing digit is $9 - 8$. Missing digit: **1**

Try it: Which digit goes in the box so that $7\square2$ is divisible by 9?

check: **9**

Skill 3 — When two repeating events meet again

Rule: Two events that repeat every m and every n steps meet again at the **least common multiple**.

List multiples of each and take the first number on both lists. Or use the shared factor: $m \times n \div$ (factor they share).

Multiplying $m \times n$ always gives *a* meeting time, but usually not the *first* one.

Example: Buses leave every 4 minutes and every 6 minutes, starting together. When do they next leave together?
Step 1. I want the first time that is a multiple of both gaps, so I list multiples of each and look for the first number they share.
Step 2. Multiples of 4: 4, 8, 12. Multiples of 6: 6, 12. They share 12, which is also $4 \times 6 \div 2$ because 2 is the factor they share. Next together: **12** minutes

Try it: Two trams leave every 9 minutes and every 15 minutes, starting together. When do they next leave together?

check: **45** minutes

Skill 4 — Reading a number from its prime factors

Rule: Split a number until only primes are left. That list is the number's building plan.

Multiply the whole list back and you get the number; any group you can pull out of the list is a factor of it.

So dividing the number by the small primes already found leaves the remaining prime factor behind.

Example: Write 84 as a product of primes and give its largest prime factor.
Step 1. To find prime factors I keep splitting off the smallest prime that divides, and the leftover at the end is the largest one.
Step 2. $84 = 2 \times 42 = 2 \times 2 \times 21 = 2 \times 2 \times 3 \times 7$, so dividing out $2 \times 2 \times 3$ leaves $84 \div 12$. Largest prime factor: **7**

Try it: Write 165 as a product of primes and give its largest prime factor.

check: **11**